Chapter - 2

Classical Crytography Techniques

Some common terminologies

- **Plain Text** - The original message or data that is fed into the algorithm as input.
- **Cipher Text** - This is the scrambled message produced as output. It depends on the plaintext and the secret key.
- **Cryptography** - The science of securing information by transforming it into unreadable ciphertext
- **Crypt Analysis** - Method of deciphering w/o knowing the key
- **Cryptology** - Combination of cryptology and crypto analysis

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Types of cryptography

3 types based on **operation**

Substitution	Transposition
❖ Caesar Cipher	❖ Rail Fence
❖ Monoalphabetic Cipher	❖ Row Column Transposition
❖ Playfair Cipher	
❖ Hill Cipher	
❖ Polyalphabetic Ciphers	
❖ One-Time Pad	

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1. **Substitution Cypher**

- Letters of plain text is replaced by letters, numbers or symbols.
- Caesar Cypher is a popular substitution cipher.
- a,b,c,...,z == > 0,1,2,...,25
- Encryption formula
- > C = (P + K) (mod 26)
  > P - Plain text value, K - Shift key
- Decryption formula

- > P = (C - K) (mod 26)

2. **Transposition Cipher**
3. **Product Cipher**

2 types based on **number of keys**

1. **Single key** / **Private Key** / **Symmetric Key**
2. **Two keys** / **Public Key** / **Asymmetric Key**

2 types on size of **plain text**

1. **Block**
2. **Stream**

Play-Fair Cipher

Rules

5X5 matrix for the word CRYPTOGRAPHY

C	R	Y	P	T
O	G	A	H	B
D	E	F	I	K
L	M	N	Q	S
U	V	W	X	Z

- Place the letters one by one in the 5X5 matrix.
- It has 25! possible keys (if keys are not repeated) i.e. 2^{84} and if keys are identical then 2^{79}
- After all the letters of the word is filled, place the letters in alphabetical order.
- Do not repeat any letters.
- If letters of a pair are same, then add X,Y or Z in between those letters.
- If letters appear on the same colomn, replace them with the letters immediate below with wrapping.
- If letters appear on the same row, replace them with the letter to the immediate right with wrapping (according to the direction)
- If letters are not on the same row or colomn, replace them with the letters on the same row respectively but at the other pair of corners of a rectangle by the original pair.

Question 1

Key - **"Cryptography class spring"** Plain text- **"Hello find the solution"**

Answer:

C	R	Y	P	T
O	G	A	H	L

C	R	Y	P	T
S	I	N	B	D
E	F	K	M	Q
U	V	W	X	Z

- Plain text pairs : HE LX LO FI ND TH ES OL UT IO NZ
- Cipher Text pairs: OM HZ GA VR BS PL UC GA CZ SG DW

Question 2

Key - **"Keyword"** Plain text - **"Why don't you"**, **"Come to the window"**, **"The big wheel"**

Answer:

K	E	Y	W	O
R	D	A	B	C
G	H	I	K	L
M	N	P	Q	S
T	U	V	X	Z

WH YD ON TY OU
EK EA ES VK EZ

CO ME TO TH EW IN DO WX
TH EB IG WH EE LX

TH EB IG WH EX EL
UG WD KL EK WU OH

Question 3

Key - **"Monarchy"** Plain text - **"I am at college right now"**

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

IA MA TC OL LE GE RI GH TN OW
BS OR LD MP UL IF AK FY QR NV

Polyalphabetic Cipher

- This is another approach to improve security by using multiple cipher alphabets called as polyalphabetic substitution cipher.
- It repeats the key from the beginning once after the end of the key is reached.
- The simplest cipher under this is Vigenere Cipher which is effectively multiple ceaser ciphers.
- The key is multiple letters long.
- Decryption in Vigenere Cipher simply works in reverse.
- The same plain text charecter is encrpted to different.

k = k1 k2 .... kd

Key	Plain Text	Cipher Text
C A T C A T C	C O L L E G E	E O E N E Z G
2 0 19 2 0 19 2	2 14 11 11 4 6 4	

Question 1

Key - **D E C E P T I V E** Plain Text - **"We are discovered, save yourself"**

D E C E P T I V E 3 4 2 4 15 19 8 21 4

Cipher Text - Z I C V T W Q N G R Z G V T W A V Z H C Q L G L M G J

A	B	C	D	E	F	G	H	I	J	K
0	1	2	3	4	5	6	7	8	9	10

L	M	N	O	P	Q	R	S	T	U
11	12	13	14	15	16	17	18	19	20

V	W	X	Y	Z
21	22	23	24	25

Security of Vigenere Cipher

- If a Vigenere Cipher is suspected, then determine the length of the keyword
- If two identical sequences of a plain text letters occur at a distance, i.e an integer multiple of the keyword length, which generates identical cipher text sequences.
- To find a solution against the know frequency using the repeated key, another approach called Auto Key System can be used.
- In Auto Key System a keyword is concantenated with the plain text to provide a running key.
- Consider the encryption according to the order of the plain text.

Vernam Pad

> Encryption $C_i = (p_i + k_i) \pmod{26}$ Decryption $p_i = (C_i - k_i) \pmod{26}$

$C_i = P_i$ XOR k_i

X	0	R
0	0	0
0	1	1
1	0	1
1	1	0

ASCII Value (A - Z) = (65 - 90) (a - z) = (97 - 122)

Question 1

Plain Text - C A T Key - A A A

Solution

ASCII - C A T = 676584 Binary - 01000011 01000001 01010100

ASCII - A A A = 656565 Binary - 01000001 01000001 01000001

$C_i = P_i$ XOR k_i C_i = 01000011 01000001 01010100 + 01000001 01000001 01000001 C_i = 00000011 00000000 01010101 C_i = 2021

- In theory it is 100% secure
- **$C_i = P_i$ XOR k_i**
- Key lengh has to be equal to the plain text length and should be purely random (generated from nature).
- Random keys are pure random keys that cannot be genrated by a computer.

Transposition / Permutation Cipher

These hide messages by rearranging the letter order w/o altering the actual lettrs used.

1. **Rail Fence Cipher**
    - Write message letters out diagonally over a no. of rows and then read off cipher row by row.
    - Do not fill empty spaces.
    - eg : "meet me after the toga party"
      m e m a t r h t g p r y e t e f e t e o a a t Cipher Text : MEMATRHTGPRYETEFETEOAAT

2. **Columnar Transposition Cipher**

- Plain Text - "Attack postponed until 2am"
- Key = 4 3 2 1 5 6 7
- Take the least key value column
- C.T - aptm ttna tsuo aodw coix knly petz
- This is column wise transposition

4	3	2	1	5	6	7
a	t	t	a	c	k	p
o	s	t	p	o	n	e
d	u	n	t	i	l	t
w	o	a	m	x	y	z

- For row wise transposition

4	a	o	d
3	t	s	u
2	t	t	n
1	a	p	t
5	c	o	i
6	k	n	l
7	p	e	a

Rotor Machine

- It is an electro mechanical encryption device that encrypts letters by passing them through a set of rotating discs called rotors.
- It changes location after every key press.
- Rotors are connected internally w the help of wires
- It was used in world war 2 which was popularly known as enigma

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Steganography

- It is the technique of hiding secret info inside another file so that no one knows a message exits.
- **Types** : Image, audio, video and text

Question

Find the PT and key for given cipher text
YMJGTTPLFIXGDGDJWSJXYANSHJSYBWNLMYITJXSTYHTSYFNSYMJQJYYJWJ

Solution

Keep trying with different key values for few letters until it makes sense

By trial and error method to find k value if k = 1 Y = (P-1) (mod 26) = Z; M = N; J = K; G = H

if k = 2 Y = A; M = O; J = L; G = J

Shift key is 5 (reverse alphabetic order)

Question

P.T is "abccab" has been encrypted with Vignere cipher and the cipher text is "bbccab". What is the key length?

Solution

a	b	c	c	a	b
b	b	c	c	a	b

Here, there are 5 matching columns. Counting the number of displaced columns, from beginning to the end of the matching column, answer is 6.

Question

20 17 13 9 7 16 15 18 11 2 10 12 1 14 5 19 4 6 3 8 encrypt the message using transposition technique " Markworth attacked by two persuit planes"

solution